

VTON-SCFA: A Virtual Try-On Network Based on the Semantic Constraints and Flow Alignment

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Abstract—An image-based virtual try-on system transfers an in-shop garment to the corresponding garment region of a reference person, which has huge application potential and commercial value in online clothing shopping. Existing methods have difficulty preserving garment texture and body details because of rough garment alignment and imperfect detail-retention strategies. To address this problem, we propose a virtual try-on network based on semantic constraints and flow alignment. The key idea of the framework is as follows: 1) a global-local semantic predictor (GLSP) is proposed to generate a reasonable target semantic map, which clearly guides the correct alignment of the in-shop garment with the body and the generation of try-on result; and 2) a novel appearance flow-based garment alignment network (AFGAN) is proposed to align the in-shop garment with the body, which is important to preserve maximum garment detail and ensure natural and realistic warping; and 3) we propose a synthesis strategy to integrate the aligned garment and the human body to preserve maximum body detail for generating a realistic result and preventing cross-occlusion and pixel confusion between different body parts. Experiments on the existing benchmark dataset demonstrate that the proposed method achieves the best performance on qualitative and quantitative experiments among the state-of-the-art virtual try-on techniques.

Index Terms—Virtual try-on, human parsing, semantic constraint, flow aligning.

I. INTRODUCTION

WITH the gradual popularity of online clothing shopping, 2D image-based virtual try-ons have become an effective way for consumers to experience try-ons online, which attracting much attention from researchers in recent years. The aim of a virtual try-on is to change the garment worn on the

reference person to a given in-shop garment while retaining the original features and details as much as possible. Although previous methods have achieved effective garment replacement, the detailed processing of garments and bodies is still a challenge due to the complex process of garment alignment and the preservation of minute body details.

Existing virtual try-on tasks can be divided into 3D model reconstruction-based [1]–[7] and 2D image-based [8]–[12] methods. The 3D model reconstruction-based method requires the additional reconstruction of garments and bodies; therefore, it needs a huge computational cost and high precision of the reconstructed model. In summary, 3D model reconstruction-based methods cannot be applied widely at present. In contrast, the 2D image-based method relies on a convolutional neural network (CNN) that requires only light image processing to achieve realistic results, which can be carried out on mobile devices and lightweight processors.

In previous works, there have been many classical studies on 2D image-based virtual try-ons. Yuan *et al.* [13] propose a virtual clothing try-on system, which automatically customizes an invisible (or partially visible) avatar based on the user's body size and skin color. It is used for proper clothing fitting, alignment, and clothing simulation in the virtual try-on system. As an initial attempt, it provides a novel idea for later studies. Han *et al.* [8] propose a two-stage try-on framework, VITON. First, a rough result is generated by an in-shop garment and human body; then, a refinement network is used to optimize the generated garment part. It provides a CNN-based research route and technique for researchers; however, there are a few flaws in preserving the detail of the garments and the human body. To solve the problem, Wang *et al.* [9] propose a toward characteristic preserving try-on framework, CP-VTON. First, it warps the in-shop garment by geometric matching; then, a try-on module is designed to synthesize the garment and body. It improves the loss of garment detail according to the special fusion mechanism. However, this approach causes some new problems, such as arms being covered by garments. To make the result more realistic, Yang *et al.* [14] propose a multistage pipeline, ACGPN, which optimizes the result by refining the warped garment and adding the generative adversarial nets (GAN) mechanism [15].

However, these methods [8], [9], [14], [16]–[19] warp an in-shop garment for aligning with the body by geometric matching, that is, a thin-plate spline (TPS) transformation [20]. It achieves aligned garments by learning a few control points, which is the most fundamental cause of the unnatural warping of garments. If deformation is not constrained, the in-shop

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Our code has been publicly available at <https://github.com/Du-Ceh/VTON-SCFA>.

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garment will be warped excessively. The convolution operation also cannot handle large spatial deformation well [21], except the color and shape; therefore, it is difficult to align garments well while maintaining the characteristics of garments (e.g., textures, logos, and embroidery).

Furthermore, in terms of body detail generation, it is not easy to obtain a realistic result in the generation of arm images (including fingers) because current methods lack a useful strategy for effective arm restoration. Moreover, with a pose with a crossed arm and garment, the garment will cover the arm in the synthesis stage, and the try-on effect will be ruined.

Most universal requirements of the virtual try-on task can be summarized as the following important challenges: 1) the invariance of the human body (e.g., head, trousers, and hands) needs to be kept fully after try-on, 2) the characteristics of in-shop garments (e.g., textures, logos, and embroidery) cannot be lost after try-on, and 3) the changed part of the human body (arms and neck) should be restored after try-on.

To realize the abovementioned requirements for obtaining a better result, we propose a three-stage virtual try-on framework. First, we design a semantic map predictor that fuses global and local features to explore the distribution of garments on the human body. The predictor emphasizes the arm priority of the semantic layer and highlights the arm position through the introduction of an arm representation map (ARM). Moreover, the neck semantic map is extended in original dataset to remove the generation limitation of the collar type. Second, we propose a novel garment-alignment network that warps the in-shop garment based on the predicted garment semantic map to align the in-shop garment with the body while keeping as many garment details as possible. Meanwhile, we eliminate the traditional TPS algorithm and learn an appearance flow transformation for garment alignment, which helps to obtain a stable and realistic alignment result. Third, the aligned in-shop garment and body invariance are integrated to generate the final try-on result. To keep the maximum detail of the arms, we design a synthesis strategy to effectively assist the generation of realistic results. The main contributions of the paper are summarized as follows:

- 1) We design the GLSP to predict the target semantic map for guiding garment alignment and the generation of try-on images. The GLSP strengthens the distribution rationality of the predicted semantic map by the global and local information of the global-local block's fusion characteristics.
- 2) We propose a novel AFGAN that estimates the appearance flow to warp in-shop garments for alignment with the body. The AFGAN enhances the warping naturalness of in-shop garments using a feature-mapping structure that can effectively preserve the maximum detail of garments.
- 3) We design a synthesis strategy to keep the arm and body invariant for generating a more realistic result. It enriches the details of the generated result and reduces the image noise to narrow the gap with the real scene.
- 4) The dataset is expanded to generate a higher-quality result. The arm representation map and neck semantic map are introduced into original dataset to optimize the predicted

semantic map, and the neck semantic map frees up the constraint of the generation of collar type.

The remainder of the paper is organized as follows: Section II is related work. Section III describes the proposed virtual try-on framework in detail. Section IV reports the experiment result. Section V discusses the existing virtual try-on work. Section VI makes a brief conclusion.

II. RELATED WORK

The technology of virtual try-on mainly includes three key issues: conditional image synthesis, human parsing and understanding, and virtual try-on methods. The related works may derive from other intelligent application studies of clothing [22]–[25]. The detailed works should be given as following.

A. Conditional Image Synthesis

The contribution of GAN [15], [26] is remarkable in the region of image generation, and GAN has a good effect on fashion image processing. Conditional GAN (CGAN) [16], [27]–[29] makes the generated image controllable according to the condition information, and the generated result of virtual try-on is based on that idea. Jetchev *et al.* [21] propose the conditional analogy GAN (CA-GAN), which defines the virtual try-on task as an image analogy problem. It adds a cyclic consistency loss function to improve the image quality. However, it loses many image details because the convolution operation cannot handle the large spatial deformation. Therefore, it is difficult to achieve the good performance of virtual try-on only with GAN. Cui *et al.* [30] propose an end-to-end virtual garment display approach to render the sketch and fabric of garments. Lee *et al.* [10] introduced an adversarial mechanism into the network to optimize the alignment degree of the garment, which made the generated result more reasonable. In our work, only the missing information of the arm and neck needs to be generated due to the special aligning method; thus, we use sufficient condition information to achieve the high-quality try-on result.

B. Human Parsing and Understanding

The generation of human semantic segmentation [31]–[33] and the estimation of human pose [34]–[36] are important human-centric image research. Human semantic maps can provide any desired image part for the experiment. The body's pose can be a warping contour for the pose of in-shop garments and arms. Long *et al.* [37] first proposed a CNN-based method called FCN for semantic segmentation. Gong *et al.* [31] propose a benchmark Look into Person (LIP) that constitutes a significant advance in terms of the target diversity. In addition, it studies self-supervised structure-sensitive learning for body parsing and body-pose estimation. Chen *et al.* [38] propose a semantic segmentation network, that combines deep convolutional neural networks (DCNNs) and dense conditional random fields (DenseCRFs); in this way, the problem of low position accuracy is solved. Cao *et al.* [36] propose a bottom-up method that realizes real-time pose estimation. It provides the basis for a virtual try-on algorithm to operate in real time. In our work, both

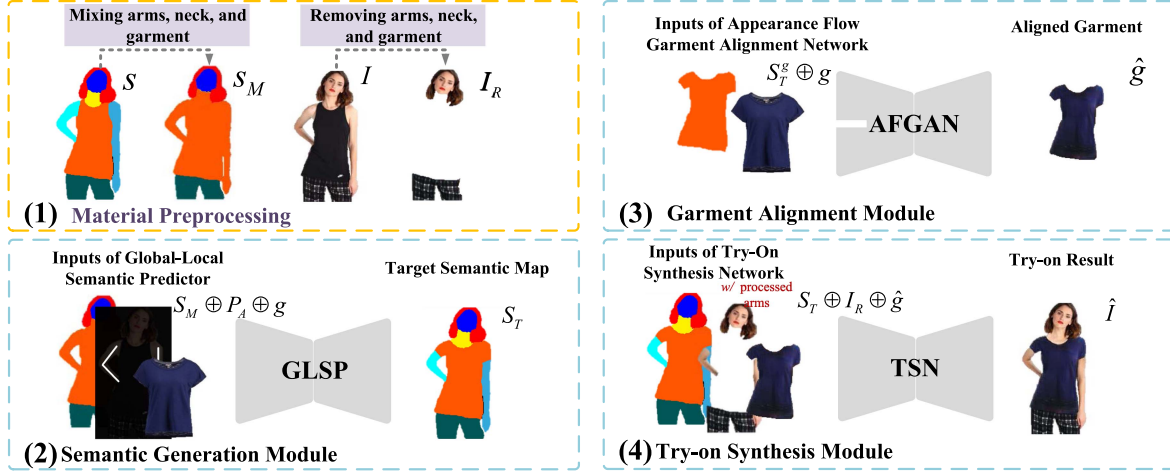


Fig. 1. Overview of the proposed virtual try-on framework $\mathcal{M}$. The network architecture consists of three main modules: 1) semantic generation module, 2) garment alignment module, and 3) try-on synthesis module. The execution order is from (1) to (4), and (1) shows material preprocessing. First, the arms, neck, and garment layers in the reference semantic map S are mixed together to obtain the semantic map S_M to be resegmented. Meanwhile, the arms, neck, and garment in the reference person I are removed to obtain the body invariant I_R ; (2) shows target semantic map generation, which uses GLSP to parse the in-shop garment g and arm representation map P_A in the input conditions to resegment S_M for predicting the target semantic map S_T ; (3) shows garment alignment, which uses AFGAN to estimate appearance flow $\mathcal{F}$ for warping g according to garment semantic map S_T^g , then the module obtains the aligned garment $\hat{g}$; (4) shows try-on result synthesis, which uses a try-on synthesis network (TSN) to integrate I_R (the processed arms is attached to the I_R) and $\hat{g}$ according to the garment semantic map S_T to obtain the virtual try-on result $\hat{I}$. $\oplus$ denotes connection in the channel direction.

pieces of information are necessary to generate virtual try-on results.

C. Virtual Try-On Methods

Current virtual try-on methods are divided into 3D model reconstruction-based [1]–[7] and 2D image-based [8]–[12] methods. Mir *et al.* [6] proposed a method to transfer the texture of garment images (front and back) to 3D garments worn on the SMPL model [39]. The experimental results indicate that the method is more accurate and faster than the picture deformation and picture-to-image transfer networks based on the TPS. Minar *et al.* [18] propose a two-stage virtual try-on framework, which imposes a limit on the TPS to control the excessive warpage of the garment. Yang *et al.* [14] propose a three-part module based on the adaptive content generation retention network that adaptively determines which part of the image is supposed to be retained, and it greatly reduces the blurred detail in the generated result. However, the handling of the arm and garment is still rough, which produces unsatisfactory results. We solve the basic garment deformation problem to generate a high-quality virtual try-on result.

III. METHODOLOGY

In this section, we first illustrate the general structure of GLSP to predict a target semantic map with the shape of an in-shop garment, which is used to guide the garment alignment and result generation. Second, we explain the general structure of the AFGAN to warp the in-shop garment, which is used to align the garment corresponding to the body's pose. Third, we explain the synthesis strategy and its specific implementation details.

Given an in-shop garment $g \in R^{3 \times H \times W}$ and a reference person $I \in R^{3 \times H \times W}$. The goal of the virtual try-on is to naturally transfer the in-shop garment g to the garment region corresponding to the reference person I to generate the try-on result $\hat{I} \in R^{3 \times H \times W}$. Our virtual try-on pipeline $\mathcal{M}$ can be expressed as follows:

$$\hat{I} = \mathcal{M} \langle g, I \rangle \quad (1)$$

As shown in Fig. 1, the framework consists of three main modules: 1) **Semantic Generation Module (SGM)**: the in-shop garment g , the arm representation map $P_A \in R^{1 \times H \times W}$, and the mixed semantic map $S_M \in R^{21 \times H \times W}$ are input into GLSP to generate the target semantic map $S_T \in R^{21 \times H \times W}$, 2) **Garment Alignment Module (GAM)**: the in-shop garment g and the predicted garment semantic map $S_T^g \in R^{1 \times H \times W}$ are input into AFGAN to estimate appearance flow $\mathcal{F} \in R^{2 \times H \times W}$ for warping g , and 3) **Try-on Synthesis Module (TSM)**: the body invariant $I_R \in R^{3 \times H \times W}$, the aligned garment $\hat{g}$, and the predicted target semantic map S_T are input into TSN to generate the virtual try-on result $\hat{I}$.

A. Semantic Generation Module

In VITON [8] and CP-VTON [9], the person shape map (human body area is filled with white pixels) is used to provide the guidance of image generation, but the person shape map only provides the rough outer contour, and the internal organizational differences of body are ignored, which leads to problems such as pixel confusion and arm occlusion by the garment. Thus, ACGPN [14] is proposed to provide a new idea that uses the semantic map to guide the generation of try-on images. However,

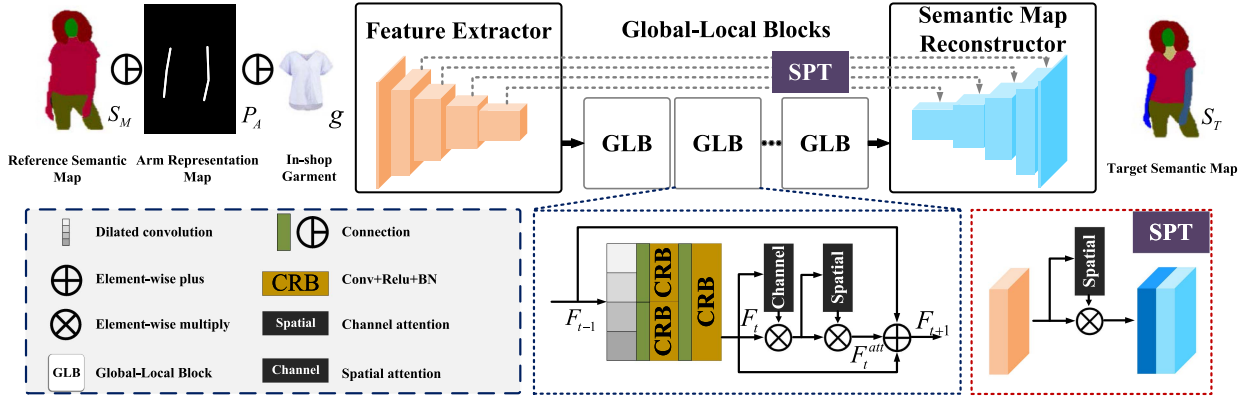


Fig. 2. Overview of the semantic generation module. We use a feature extractor to extract the feature of the input condition information and the semantic map reconstructor to generate the target semantic map S_T . We join several GLBs between the feature extractor and the semantic map reconstructor in GLSP, and the spatial attention module (SPT) is joined into the jump connection structure to refine and enhance the fusion of local and global features.

it is unavoidable that a large number of semantic maps with errors (e.g., collar type and sleeve length mismatching the in-shop garment) are generated, which causes the result to be poor.

To solve the abovementioned problems for predicting the correct target semantic map as much as possible, we design an efficient semantic generation pipeline containing the semantic predictor and the technological process.

Note that there is no neck semantic map in the reference semantic map S ; thus, we add the neck semantic map to S to solve the range restriction of the garment semantic map.

First, the semantic map except for the neck, garment, and arms is unchanged before and after try-on. We mix the three parts into one layer as the mixed semantic map S_M , and then let GLSP learn the resegmentation ability of the mixed area. In addition, the in-shop garment g is used as one of the inputs to provide a shape basis of the garment layer for resegmentation.

1) *Arm Representation Map*: In the changeable poses of the human, it is inevitable in which there will be a pose that arms and garments are crossed and overlapped. The kind of pose causes the garment semantic map to cover the arm semantic map. To solve the problem, we design an arm representation map P_A that connects the shoulder points, elbow points, and wrist points with a white line on the single channel black map.

When the arm representation map is the input of GLSP, the arms' position is highlighted accurately, which effectively reminds GLSP where the arm representation map exists that the arm semantic map should be generated. Therefore, the problem of arm occlusion by the garment is solved effectively.

2) *Global-Local Semantic Predictor*: As shown in Fig. 2, we design the GLSP to generate a target semantic map S_T . Previous works (e.g., ACGPN [14]) adopt U-Net [40] as the semantic predictor. Although U-Net can fulfill the prediction task, there are many errors in the predicted semantic map (e.g., the collar type and sleeve length are mismatched with the in-shop garment) due to the lack of global and local information relation; that is, there is no structure to handle the local and global relationship in U-Net. This can cause the arm and garment characteristics to be missing further. Therefore, we improve U-Net by

introducing the global-local block (GLB) to fuse the global and local feature maps of input to improve the accuracy of semantic segmentation.

The GLSP consists of a feature extractor, a semantic map reconstructor, and several GLBs between the feature extractor and the semantic map reconstructor. Moreover, a jump connection structure with the spatial attention mechanism (it adds a spatial attention module to the previous U-Net's jump connection part, see SPT in Fig. 2) is introduced into GLSP to further enhance the feature fusion. The feature extractor is composed of five convolutional operations, and the semantic map reconstructor is composed of five deconvolution operations. The detailed structure of the GLB is shown in the blue dotted box in Fig. 2. The GLB uses the dilated convolution of four ratios to capture the global and local spatial information of features and introduces the attention mechanism to refine and enhance the fusion of local and global features.

First, the input feature F_{t-1} is extracted in parallel with n -ratio dilated convolution. Then, every two of the n extracted features are divided into groups for connection in the channel direction, and then the dimensions of the connected features are halved by convolution again. Finally, the obtained feature is concatenated and convolved once more to gain the feature map F_t . The formula is defined as follows:

$$F_t^{grp} = Re(f^{1 \times 1}(\Phi_{t-1}^n(F_{t-1}); \Phi_{t-1}^{n+1}(F_{t-1}))) \quad (2)$$

$$F_t = Re(f^{1 \times 1}(F_t^1; F_t^2)) \quad (3)$$

where t is the different stages of feature processing, and Re denotes the ReLU activation function. grp denotes the feature group index, $grp \in \{1, 2\}$. Φ_{t-1}^n denotes the dilated convolution; n is its dilated ratio, and $n \in \{1, 3\}$; $\cdot$ denotes the connection in channel direction. $f^{1 \times 1}$ denotes the convolution operation with a filter size of 1×1 to fuse the multiscale feature.

Furthermore, we use the channel and spatial attention modules from [41] to enhance and refine the global and local feature map

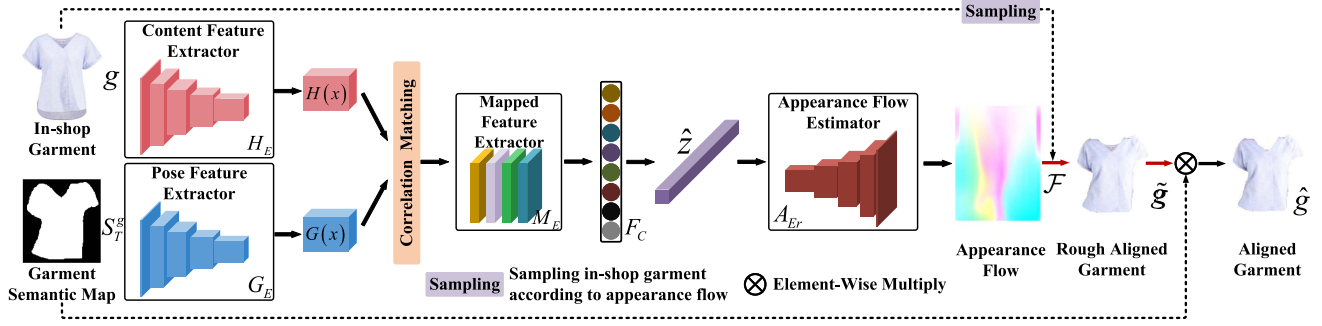


Fig. 3. Overview of the garment alignment module. We use a content feature extractor H_E to extract the feature of the in-shop garment g , and a pose feature extractor G_E is used to extract the feature of the garment semantic map S_T^g . A correlation matching module is used to find the mapping between features $H(g) \in \mathbb{R}^{C \times 16 \times 12}$ and $G(S_T^g) \in \mathbb{R}^{C \times 16 \times 12}$. A mapped feature extractor M_E and a full connection layer F_C are used to refine the mapped feature $\mathcal{T}_D \in \mathbb{R}^{(16 \times 12) \times 16 \times 12}$ to obtain the vector $\hat{z}$ further. Finally, an appearance flow estimator A_{Er} is used to estimate the appearance flow $\mathcal{F}$ for the warping garment. The redundant area of a roughly aligned garment $\tilde{g}$ is eliminated through its product $\otimes$ and garment semantic map S_T^g . Thus, the final aligned garment $\hat{g}$ is obtained.

F_t . The formula is defined as follows:

$$M^c(F) = \sigma \left(W_1 \left(W_0 \left(F_{avg}^c \right) \right) + W_1 \left(W_0 \left(F_{max}^c \right) \right) \right) \quad (4)$$

$$M^s(F) = \sigma \left(f^{3 \times 3} \left([F_{avg}^s; F_{max}^s] \right) \right) \quad (5)$$

where $M^c(F)$ and $M^s(F)$ are feature maps extracted by channel and spatial attention, respectively. σ denotes the sigmoid function; W_0 and W_1 are multilayer perceptron weights, and F_{avg}^c and F_{max}^c denote the average-pooling and max-pooling features, respectively. $f^{3 \times 3}$ denotes the convolution operation with a filter size of 3×3 .

Finally, we add a jump connection structure such as ResNet [42] into GLB. F_{t-1} , F_t , and F_t^{att} are added element-wise to generate the output feature F_{t+1} of GLB further. It can be described as follows:

$$F_t^{att} = M^s(M^c(F_t)) \quad (6)$$

$$F_{t+1} = F_{t-1} + F_t + F_t^{att} \quad (7)$$

B. Garment Alignment Module

Current virtual try-on methods, such as VITON [8], CP-VTON [9], and ACGPN [14], use the TPS [20] to warp an in-shop garment to align the in-shop garment with the body. It works by calculating control points and overlapping the control points with predetermined points on the garment image by 2D interpolation. Obviously, the aligned garment gained by a few control points is subject to unnatural and uneven warping because the TPS warps a large area of the garment mechanically to achieve alignment. Therefore, to achieve the adaptive position offset for each pixel, we propose the appearance flow garment alignment network (AFGAN) to estimate the dense appearance flow [43] $\mathcal{F} \in \mathbb{R}^{2 \times H \times W}$ corresponding to changes in garment type. The estimated appearance flow uses the variation and correlation between pixels in the *in-shop garment - ground truth garment* pair $(g, \bar{g})$ to learn the correspondence that exists between g and $\bar{g}$ to learn the alignment model between g and $\bar{g}$ to obtain aligned garment $\hat{g}$.

1) *Appearance Flow Garment Alignment Network*: As shown in Fig. 3, AFGAN consists of three encoders, a correlation matching module, and a decoder. Two encoders, the content feature extractor H_E and the pose feature extractor G_E , are the same, and both consist of five layers of convolution. In the correlation matching module (CMM), we use the in-shop garment's feature $H(g) \in \mathbb{R}^{C \times 16 \times 12}$ and the garment semantic map's feature $G(S_T^g) \in \mathbb{R}^{C \times 16 \times 12}$ from the content feature extractor H_E and the pose feature extractor G_E , respectively, to map both to the common domain $\mathcal{T}_D \in \mathbb{R}^{(16 \times 12) \times 16 \times 12}$ so that AFGAN can learn the reliable content correspondence of the input pair (g, S_T^g) in $\mathcal{T}_D$.

$$\mathcal{T}_D(u,v,w) = H(g)_{(u,v)}^T G(S_T^g)_{(u_w,v_w)} \quad (8)$$

where $\mathbb{T}$ denotes matrix transpose; (u, v) and (u_w, v_w) denote the individual feature positions in the 16×12 feature maps, and $w = 16(v_w - 1) + u_w$ is an auxiliary index variable for (u_w, v_w) .

Another encoder, the mapped feature extractor M_E , consists of four layers of convolution. It is used to increase the channel depth of $\mathcal{T}_D$ further, and there is a 786-dimensional fully connected layer F_C at the end of M_E to yield a vector $\hat{z}$ with the latent appearance-flow information. The appearance-flow estimator A_{Er} consists of five layers of deconvolution operations, which decode the vector $\hat{z}$ to estimate the desired dense appearance flow $\mathcal{F}$ for the warping garment.

2) *Flow Correction*: In practice, we find that the aligned garment image $\tilde{g}$ obtained by the appearance flow can exhibit unnatural warping (i.e., excessive and irregular deformations) because the high flexibility is not corrected positively during training. As shown in Fig. 4(c), to address the problem, we introduce a regularization correction loss $\mathcal{L}_c$ of the appearance-flow coordinate to prevent local anomalies in the estimated dense appearance flow $\mathcal{F}$. By imposing a balanced restriction between each flow coordinate and its adjacent coordinate, it effectively avoids the local overwarping caused by the excessive distance between some coordinates. The formula is defined

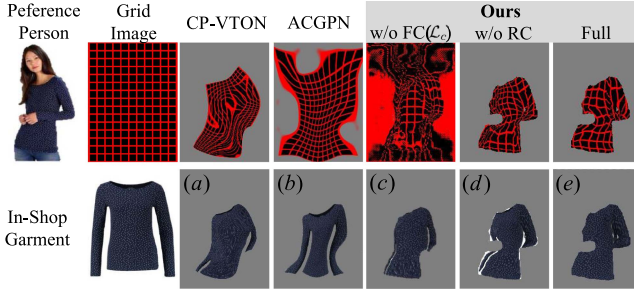


Fig. 4. Verification of the flow correction and region constraint. Column (a) is the TPS's result without any constraints (by CP-VTON [9]). Column (b) is the TPS's result with the second-order-difference constraint (by ACGPN [14]). Column (c) is the result of AFGAN without flow correction (FC, $\mathcal{L}_c$). Column (d) is the result of AFGAN without region constraint (RC). Column (e) is the result of full AFGAN.

as follows:

$$\begin{aligned} \mathcal{L}_c(\hat{p}_x, \hat{p}_y) = & \sum_{i=-1,1} \sum_x \sum_y \left| \hat{p}_x(x+i, y) - \hat{p}_x(x, y) \right| \\ & + \sum_{j=-1,1} \sum_x \sum_y \left| \hat{p}_y(x, y+j) - \hat{p}_y(x, y) \right| \end{aligned} \quad (9)$$

where $\hat{p}_x$ and $\hat{p}_y$ are the coordinates of x and y in $\mathcal{F}$, and the absolute difference $|a - b|$ is the distance between two adjacent nodes, a and b . $\mathcal{L}_c$ totals the distance of a certain point (x, y) and its adjacent points $((x+1, y), (x-1, y), (x, y+1), (x, y-1))$.

3) *Region Constraint*: Although the regularization correction loss $\mathcal{L}_c$ of the appearance flow is an effective optimization method for unnatural warping, it is also a challenge that redundant pixels [pixels in place that are invisible or covered after changing pose, see Fig. 4(d)] in the in-shop garment image will be sampled by AFGAN, and it may still be a key factor in the formation of other unnatural warping phenomena (e.g., texture clutter, local squeeze). To solve this problem, we try to constrain the warping range of garments further. During the training and testing stage, we perform the product operation $\otimes$ between the predicted garment semantic map S_T^g and the sampled rough garment $\tilde{g}$ to achieve fixed removal of redundant area; then, the optimized aligned garment $\hat{g}$ should be obtained. It can be expressed as follows:

$$\hat{g} = S_T^g \otimes \tilde{g} \quad (10)$$

4) *Occlusion Handling*: In previous works, aligned garments were often occluded by the hair and arm in the process of warping, which caused areas of hair and arm to squeeze the fabric around them. However, in our work, the problem is convincingly solved by *region restriction*. As shown in Fig. 5, the hair and arm areas corresponding to the garment are adaptively removed by the predicted garment semantic map.



Fig. 5. Verification of the occlusion handling. Row 1 is the case study of hair occlusion, and row 2 is the case study of arm occlusion. Column 1 is the in-shop garment; Column 2 is the result of TPS without occlusion handling (by CP-VTON [9]); Column 4 is the result of AFGAN with occlusion handling.

C. Try-On Synthesis Module

To transfer the in-shop garment to any body with an arbitrary pose efficiently and realistically, inspired by the 'image-to-image' pattern [44], [45], we use the try-on synthesis network (TSN) to integrate the aligned garment $\hat{g}$ and the body invariance I_R to generate the desired result based on the semantic condition S_T .

1) *Synthesis Strategy for Preserving Maximum Arm Detail*: The restoration of arms after the try-on is a difficult challenge all along, and the reason is as follows: 1) the arm pose is complicated and irregular, 2) the arm shape will be changed with the change of garment, and 3) the finger part is small and difficult to generate in high-definition. Previous works adopt the adaptive generation of the whole arm area or reserve the hand area while generating other parts of the arm that cannot maximally retain the original arms, especially when the arms overlap (see Fig. 9, row 3). During the training stage, we remove the fixed length of the arms from the shoulder and use the remaining part as the arm information. Then, TSN can learn the supplementary ability of the missing part. During the testing stage, we perform product operation $\otimes$ between the original arms I^a and the predicted arm semantic map S_T^a to obtain the maximized arm area $\hat{I}^a$. Therefore, when the missing part of arms exists, it can be supplemented by TSN. When an excess of the arms exists, it can be removed by product operation, which can be expressed as follows:

$$\hat{I}^a = S_T^a \otimes I^a \quad (11)$$

Moreover, the neck area is generated by the TSN based on the predicted neck semantic map S_T^n .

2) *Try-On Synthesis Network*: As shown in Fig. 6, TSN is constructed by adding a spatial attention structure (different from SPT in GLSP) to the jump connection structure of U-Net to form the attention U-Net [46]. The experiment shows that the network can make the result closer to ground truth because the attention block can refine the extracted features.

Note that the generated aligned garment $\hat{g}$ can be used directly for input without the extra refinement (e.g., ACGPN [14]). Finally, TSN can complete the integration of $\hat{g}$ and I_R (the

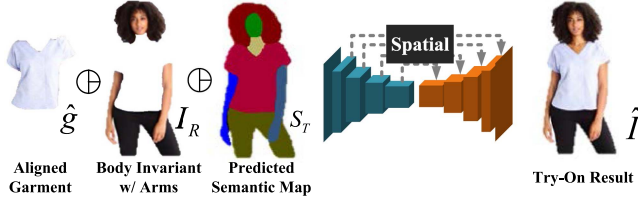


Fig. 6. Overview of try-on synthesis module. The input of TSN is aligned garment $\hat{g}$, body invariant I_R (the processed arms is attached to the I_R), and predicted semantic map S_T .

processed arms are attached to the I_R) based on predicted semantic map S_T well to generate the desired try-on result $\hat{I}$.

D. Training

During the training stage, we optimize each network separately with the different loss functions.

1) *Reconstruction Loss*: To guide the network to generate a result with a visual appearance close to ground truth, the reconstruction loss $\mathcal{L}_{rec}$ is introduced to total the pixel-level differences. $\mathcal{L}_{rec}$ is expressed as the $\mathcal{L}_1$ distance between the generated result $\hat{I}$ and ground truth $\bar{I}$.

$$\mathcal{L}_{rec}(\bar{I}, \hat{I}) = \|\hat{I} - \bar{I}\|_1 \quad (12)$$

where $\|a - b\|_1$ is the Euclidean distance between a and b .

2) *Perceptual Loss*: It is practical to extract the feature map of the generated result $\hat{I}$ and ground truth $\bar{I}$ from some layers of the pretrained VGG network [47]. The generated result is similar to ground truth in terms of visual perception by feature matching. Therefore, we introduce the perceptual loss $\mathcal{L}_{per}$, which can be expressed as:

$$\mathcal{L}_{per}(\bar{I}, \hat{I}) = \sum_{i=0}^n \alpha_i \|\phi_i(\hat{I}) - \phi_i(\bar{I})\|_1 \quad (13)$$

where $\phi_i(\cdot)$ denotes the i -th ($i = 0, 1, 2, 3, 4$) layer feature map in the pretrained VGG19 network ϕ of the input image, and α_i controls the weight of loss for each layer.

3) *Overall Loss*: In our framework, GLSP adopts the weighted cross-entropy loss $\mathcal{L}_s$ [31] to optimize the network. AFGAN adopts the reconstruction loss $\mathcal{L}_{rec}$, the perceptual loss $\mathcal{L}_{per}$, and the regularization correction loss $\mathcal{L}_c$ to optimize the network. It can be expressed as:

$$\mathcal{L}_{total}^{AFGAN} = \lambda_{rec}\mathcal{L}_{rec} + \lambda_{per}\mathcal{L}_{per} + \lambda_c\mathcal{L}_c \quad (14)$$

where λ is the trade-off hyperparameter for the corresponding loss function.

The TSN adopts the reconstruction loss $\mathcal{L}_{rec}$ and the perceptual loss $\mathcal{L}_{per}$ to optimize the network. It can be expressed as:

$$\mathcal{L}_{total}^{TSN} = \lambda_{rec}\mathcal{L}_{rec} + \lambda_{per}\mathcal{L}_{per} \quad (15)$$

IV. EXPERIMENTS

In this section, first, we introduce the experimental dataset VITON [8] and describe the implementation details of the experiment. Second, we compare qualitative and quantitative results

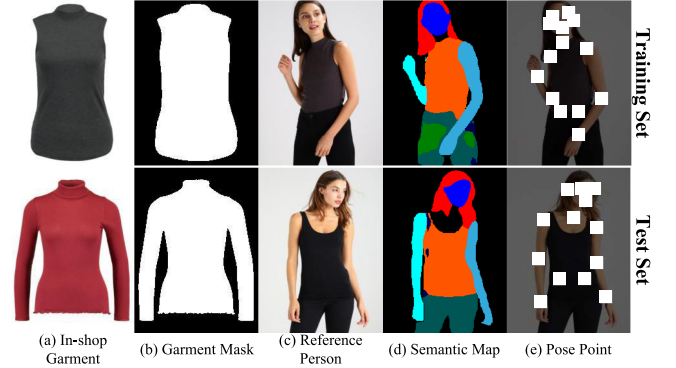


Fig. 7. An example of the training set and the test set in the VITON dataset. Row 1 is the training set, and row 2 is the test set. (a) Shows the in-shop garment. (b) Shows the garment mask. (c) Shows the reference person. (d) Shows the semantic map. (e) Shows the pose point. Note that the in-shop garments in the test set are different from the garment worn on the reference person.

with other state-of-the-art methods. Finally, we conduct ablation experiments to demonstrate the effectiveness of each module in the framework.

A. Dataset

The experimental dataset is from the VITON [8] dataset collected by Han *et al.* It contains 16,253 image groups, and each image group contains a woman image I of the front view, an in-shop garment g and its mask, a set of pose points, and an original semantic map S . Each item is defined at the size of 256×192 . The dataset contains 14,221 groups of training data and 2,032 groups of test data. As shown in Fig. 7, we show a group of training sets and a group of test sets.

B. Implementation Details

All experiments are carried out on a 1 Tesla V100 GPU with 32 GB RAM. By default, the learning rate for the generator is 0.0001, and it is reduced linearly to 0 in half of the epochs with a batch size of 4. The experiment adopts the ADAM optimizer [48], and the parameters are set as follows: $\beta_1 = 0.5$, $\beta_2 = 0.999$. In the loss function, $\lambda_{rec} = \lambda_{per} = 1$, $\lambda_c = 40$ in $\mathcal{L}_{total}^{AFGAN}$, $\lambda_{rec} = \lambda_{per} = 1$ in $\mathcal{L}_{total}^{TSN}$.

C. Qualitative Results

We visually compare our method with the existing state-of-the-art methods, CP-VTON [9], CP-VTON+ [18], and ACGPN [14].

1) *Garment Alignment Effect*: First, we conduct a garment-alignment experiment to compare the garment-warping effect. To make a fair comparison, the result of ACGPN is generated in the unrefined stage. Fig. 8 shows that CP-VTON (column 4) has occlusion problems, especially arm occlusion, and part of the fabric is squeezed by the arm and hair. The garment is excessively stretched by TPS, so the patterns and logos of the garment are extremely unnatural. CP-VTON+ (column 5) effectively improves excessive stretching by adding a loss function for restraining TPS. However, the result generated by CP-VTON+



Fig. 8. Comparison of the alignment effect with state-of-the-art try-on methods (CP-VTON, CP-VTON+, ACGPN, and our method) on the test set. AFGAN generates more naturally aligned garments based on the garments semantic map while preserving the characteristics of in-shop garments. The red dotted box is our result.

still cannot eliminate the unnaturalness; that is, the phenomenon of local overwarping still exists. ACGPN (column 6) uses the second-order difference constraint to strengthen the control of excessive warping; instead, it makes the aligned garment overly uniform, and the area of the sleeve is overly expanded so that the sleeve squeezes the surrounding fabric. Although TPS can meet the task of garment alignment, its results may still fall far short of ground truth (column 8). Our network (column 7) replaces TPS with the dense appearance flow, and it can eliminate the unreasonable warping based on the accurately predicted coordinate offset of each pixel of the in-shop garment, where the excessive deformation is controlled by the regularization correction loss $\mathcal{L}_c$, and redundant pixels are eliminated by the region constraint. The results show that our method is more effective and closer to the ground truth when aligning in-shop garments with the human body.

2) *Virtual Try-On Effect*: Then, we visually compare the virtual try-on result among CP-VTON [9], CP-VTON+[18], ACGPN [14], and our method.

The reconstruction capability of the framework is experimented first (aligned image pairs (g, I) , the in-shop garment and the garment worn on the reference person are the same). As shown in Fig. 9, the effect of CP-VTON (column 4) suffers

from arm occlusion by the garment, texture confusion, massive loss of arm and body invariance, and unnatural warping of the garment. Although CP-VTON+ (column 5) imposes a warping constraint, problems such as texture confusion due to the lack of semantic map guidance, arm occlusion by the garment, and the loss of arm detail still exist. ACGPN (column 6) improves the alignment effect by refining the garment individually, and the partial arm area is retained from the randomly truncated arm. However, the rough garment obtained by TPS and the under-sized arm retained make the generated result unsatisfactory.

Meanwhile, we compare the try-on effect among state-of-the-art methods (unaligned image pairs (g, I) , the in-shop garment, and the garment worn on the reference person are different). As shown in Fig. 10, similar to the abovementioned methods, the unnatural garment alignment and arm occlusion in the three methods are the most significant. In particular, the mismatch between the generated collar and the in-shop garment is also a weakness. Our method (column 8) effectively retains the maximum arm area through the arm synthesis strategy. The collar type and garment details are also matched and retained well based on the correct semantic map predicted by GLSP.

As shown in column 7 (in Fig. 9) and column 8 (in Fig. 10), our framework has the dual coping strategy in terms of arm



Fig. 9. Comparison of the try-on effect with state-of-the-art try-on methods (CP-VTON, CP-VTON+, ACGPN, and our method) on the test set (aligned image pairs (g, I)). The in-shop garment and the garment worn on the reference person are the same in the experiment. Our method generates more highly realistic try-on results. The red box represents the defects.

occlusion by garment. First, the arm representation map as the input of GLSP highlights the arm position. Second, the region constraint of AFGAN eliminates the redundant pixels of the arm area of the in-shop garment. As a result, arm occlusion is improved significantly. Meanwhile, the color, texture, logo, and pattern of the in-shop garment are well preserved by the appearance flow. Finally, the arm area is retained maximally through the synthesis strategy, and by the integration of TSN, our framework generates a very advantageous result.

D. Quantitative Results

We study the performance of our framework in the benchmark metrics of image quality. The experiment adopts structural similarity (SSIM) [49] to evaluate the similarity between the try-on result and ground truth through unaligned image pairs (g, I) . The inception score (IS) [50], fr chet inception distance (FID) [51], and peak signal to noise ratio (PSNR) are adopted to measure the image quality between the try-on result and ground truth with aligned image pairs (g, I) . To make a fair comparison, the experiment evaluates the quantitative results produced by different methods under the same configuration. The PSNR can be

TABLE I
QUANTITATIVE EVALUATION OF DIFFERENT METHODS ON THE TEST SET. COMPARED WITH STATE-OF-THE-ART METHODS, OUR METHOD ACHIEVES THE SUPERIOR PERFORMANCE IN IS, SSIM, PSNR, AND FID, RESPECTIVELY. PART OF THE DATA COMES FROM [19]. A LOWER FID INDICATES A BETTER EFFECT.

Network	IS [50]	SSIM [49]↑	PSNR↑	FID [51]↓
CA-GAN [21]	2.56	0.740	-	47.34
VITON [8]	2.29	0.740	-	55.71
CP-VTON [9]	2.59	0.720	21.111	24.45
CP-VTON+ [17]	2.75	0.750	-	21.04
SieveNet [52]	2.82	0.766	16.980	14.65
ACGPN [14]	2.69	0.810	23.067	16.64
DCTON [19]	2.85	0.830	-	14.82
VTON-SCFA	2.80	0.901	27.280	7.82
Real	2.88	1	N/A	0

expressed as:

$$PSNR = 10 \log_{10} \frac{255^2}{\varepsilon} \quad (16)$$

where ε denotes the mean square error (MSE) between the generated image and ground truth.



Fig. 10. Comparison of the try-on effect with state-of-the-art try-on methods (CP-VTON, CP-VTON+, ACGPN, and our method) on the test set (unaligned image pairs (g, I)). The in-shop garment and the garment worn on the reference person are different in the experiment. Our method generates more highly realistic try-on results, which simultaneously handles misalignment between the in-shop garment and the reference person, effectively preserves the characteristics of the in-shop garment, and retains clear body invariance. The red box represents the defects.

Table I shows the IS, SSIM, PSNR, and FID scores by CA-GAN [21], VITON [8], CP-VTON [9], CP-VTON+[18], SieveNet [52], ACGPN [14], and DCTON [19]. It can be seen that the parameter values of our method among the four indicators achieve the best scores. The values are more remarkable than those of the state-of-the-art DCTON network, and the metrics obtain superior image quality and generate diversity. The value of the proposed method is 8.7% higher than DCTON in terms of SSIM, 17.8% higher than ACGPN in PSNR, and 36.7% lower than that DCTON in FID, which shows the superiority of our model in dealing with garment alignment and virtual try-on.

Meanwhile, our method can obtain more advantageous results with fewer computational resources. Table II shows the time cost and the computational complexity analysis (1 Nvidia Telsa V100 GPU) from ACGPN [14], DCTON [19], and our model.

Table II illustrates that the training time of the proposed method is almost four times faster than ACGPN and DCTON, and the model parameters and the *floating-point operations per second* (FLOPS) are also smaller than ACGPN and DCTON. The speed of approximately 20 fps proves that our method can satisfy clothing shopping online with less time cost and computational complexity.

TABLE II
TIME COST AND COMPUTATIONAL COMPLEXITY ANALYSIS. THE LOWER PART OF THE TABLE IS THE PERFORMANCE OF THE THREE MODULES OF THE PROPOSED METHOD. PART OF THE DATA COMES FROM [19]

Methods	Training Time	#Params	FLOPS	FPS
ACGPN [14]	~ 40h	139M	206G	10
DCTON [19]	~ 44h	153M	194G	19
VTON-SCFA (ours)	~ 10h	135M	184G	20
SGM ($r = 3$)	~ 0.4h	68.50 M	48.42G	34
GAM	~ 7.6h	31.33 M	33.64G	131
TSM	~ 1.7h	34.89 M	101.48G	90

r denotes the number of GLBs.

E. Ablation Study

We perform ablation experiments on the three modules of the proposed method to verify their role and effectiveness.

1) *Effectiveness of the Global-Local Block (in GLSP) and Arm Representation Map in SGM*: First, we conduct ablation experiments for the number of GLBs to find the optimal number of GLBs. We use three common semantic segmentation metrics: pixel accuracy (PA), mean pixel accuracy (MPA), and mean

TABLE III
ABLATION EXPERIMENTS FOR THE NUMBER OF GLBs IN GLSP

	$r = 0$	$r = 1$	$r = 2$	$r = 3$	$r = 4$	$r = 5$	$r = 6$	$r = 7$
PA	0.93932	0.94050	0.94023	0.94071	0.94031	0.94007	0.94010	0.93984
MPA	0.68606	0.72098	0.70623	0.73262	0.73766	0.70843	0.72334	0.72192
MIoU	0.62742	0.66771	0.65227	0.67968	0.68332	0.65389	0.67170	0.66746
$r = 8$	$r = 9$	$r = 10$	$r = 11$	$r = 12$	$r = 13$	$r = 14$	$r = 15$	$r = 16$
0.93954	0.93895	0.94016	0.93967	0.94008	0.93974	0.93969	0.93958	0.93933
0.74036	0.72337	0.72782	0.74984	0.73334	0.72031	0.74694	0.70589	0.74682
0.68437	0.66539	0.67437	0.69503	0.67976	0.66485	0.69151	0.65164	0.69256

r denotes the number of GLBs.

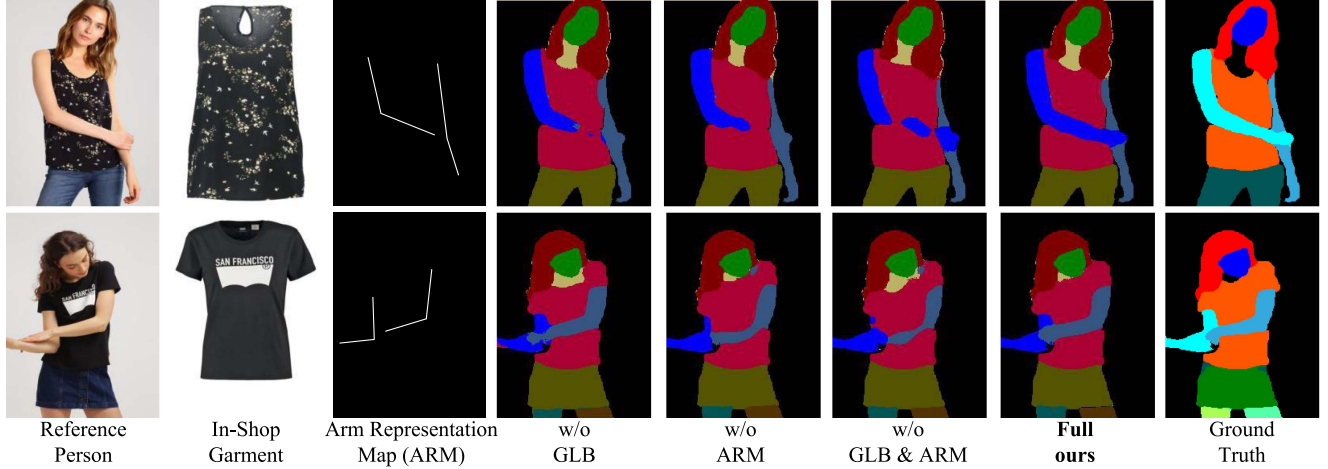


Fig. 11. Visualized ablation study on SGM. It consists of removing the GLB, removing the arm representation map (ARM), and removing the GLB & ARM.

intersection over union (MIoU). As shown in Table III, when the number r is 0 (GLSP without GLB), its effect is the worst. However, when the number r is 11, the effect reaches its maximum. The other numbers of GLBs between $r = 3$ and $r = 11$ remain relatively stable in a similar interval. We consider the size of the model to ensure real-time performance while ensuring correctness. The values of our recommended number r can be defined as 3 to 11.

Meanwhile, we apply GLSP without GLB, GLSP without the arm representation map (ARM), and GLSP without GLB and ARM to generate the semantic map. Fig. 11 demonstrates that after removing GLB from the GLSP, it has some local semantic errors, such as a generated collar shape that does not match the in-shop garment and a partial arm layer occupied by the garment layer. After removing ARM from the SGM, the mismatch of collar shape is improved, but arm occlusion still exists. After removing GLB and ARM, the same problems arise as described above, and errors in the generated arm layer and neck layer are even more unpredictable. In the full SGM ($r = 3$), the above problems are improved effectively. This demonstrates that the ARM can emphasize the priority of the arm layer in the predicted semantic map. GLB can handle the relationship of generating semantic maps between global and local wells, which may produce a more reasonable result.

Table IV shows the numerical analysis corresponding to the ablation experiments above. This proves the effectiveness of each module and the feasibility of SGM.

TABLE IV
THE ABLATION STUDY ON GLB ($r = 3$) AND ARM REPRESENTATION MAP (ARM). IT CONSISTS OF REMOVING GLB, REMOVING ARM, AND REMOVING GLB AND ARM

	w/o GLB	w/o ARM	w/o GLB & ARM	full SGM
PA	0.93932	0.92361	0.92345	0.94071
MPA	0.68606	0.68513	0.71061	0.73262
MIoU	0.62742	0.61065	0.63843	0.67968

2) *Effectiveness of the AFGAN, Flow Correction, and Region Constraint in GAM:* We verify the validity of the GAM, which consists of six parts: AFGAN without correlation matching module (CMM), GAM without flow correction (FC, $\mathcal{L}_c$), GAM without region constraint (RC), GAM without FC and RC, replacing AFGAN with TPS, and full AFGAN. Fig. 12 illustrates that the operation of removing CMM from AFGAN may result in a mismatch between the garment texture and the semantic spatial. The distribution of logos and patterns is misaligned with the human body after garments are warped. After removing FC ($\mathcal{L}_c$), the warping effect of the garment is unnatural (i.e., the local fabric of the garment will be warped excessively). After removing RC, the flow correction ensures the uniform deformation of the garment, but the boundary of the garment loses the auxiliary definition, and one side of the garment is squeezed. After removing FC and RC, the lack of boundary definition accentuates the excessive local warping of the garment. In column 6, the result of TPS is misaligned with the human body, and the

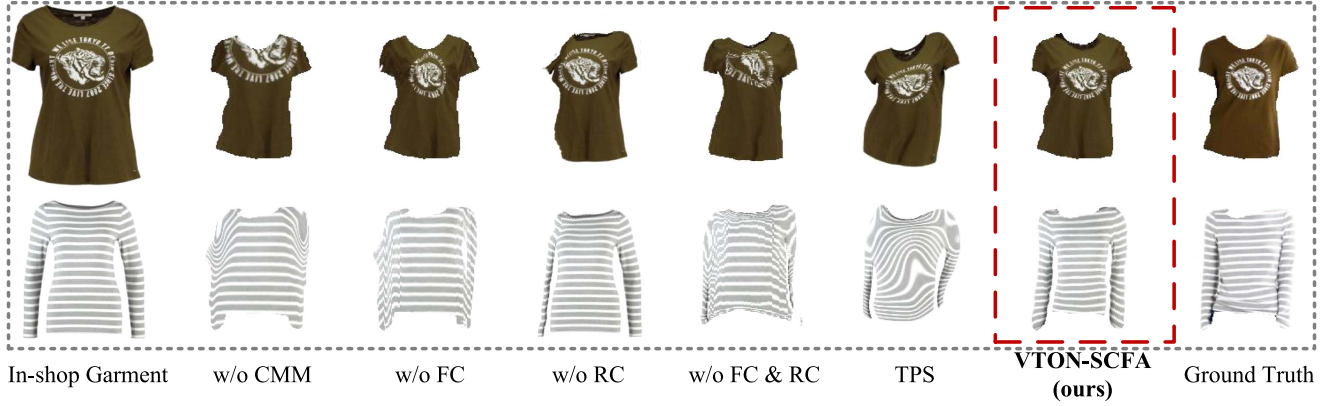


Fig. 12. Visualized ablation study on GAM. It consists of removing the correlation matching module (CMM), removing the flow correction (FC, $\mathcal{L}_c$), removing the region constraint (RC), and removing the $\mathcal{L}_c$ & RC. The 6th column is the aligning effect of TPS.

TABLE V
ABLATION STUDIES OF GAM, LOWER FID INDICATES BETTER ALIGNMENT EFFECT. FULL GAM ACHIEVES SUPERIOR PERFORMANCE IN FID

Different configurations of the AFGAN	FID [51]↓
Replacing AFGAN with TPS.	50.86236
w/o correlation matching module.	38.09899
w/o region constraint.	61.97408
w/o regularization correction loss $\mathcal{L}_c$.	35.24539
w/o correlation matching module & $\mathcal{L}_c$.	39.92060
Full GAM	23.74158

TABLE VI
ABLATION STUDIES OF TSM. IT CONSISTS OF TRY-ON SYNTHESIS WITH TPS, REMOVING THE SEMANTIC MAP (SM), REMOVING THE SYNTHESIS STRATEGY (SS), REMOVING THE SPATIAL ATTENTION (SA) STRUCTURE, AND FULL TSM

	IS [50]	SSIM [49]↑	PSNR↑	FID [51]↓	LPIPS [53]↓
w/ TPS	2.7003±0.1268	0.8908	25.9235	11.1756	0.094
w/o SM	2.7922±0.1362	0.8887	25.7715	8.4264	0.091
w/o SS	2.8067±0.1327	0.8843	25.8302	7.8950	0.088
w/o SA	2.8119±0.1329	0.8943	26.0788	8.4063	0.083
Full TSM	2.8025±0.1240	0.9015	27.2800	7.8201	0.080

Lower FID or LPIPS indicate a better effect.

excessive warping without constraints also makes the alignment effect not extremely smooth. Therefore, the full GAM has the best effect in terms of garment alignment, and it is the closest to the ground truth.

In addition, we use FID to evaluate the perception similarity between the aligned garment $\hat{g}$ and ground truth $\bar{g}$ to prove the superiority of our method. Table V demonstrates the conclusion mentioned-above; the FID of the full GAM is decreased by 53.3% from 50.86236 (TPS) to 23.74158.

3) *Effectiveness of the Synthesis Strategy and TSN in TSM:* We add a new metric, *learned perceptual image patch similarity* (LPIPS) [53], to measure the effectiveness of TSN. Fig. 13 and Table VI show the effect of five variants: 1) try-on synthesis with TPS, 2) try-on synthesis without a predicted semantic map, 3) try-on synthesis without an arm image (synthesis strategy), 4) TSN without a spatial attention structure, and 5) full TSM.



Fig. 13. Visualized ablation study on TSM. It consists of replacing AFGAN with TPS, removing the semantic map, removing the synthesis strategy, removing the spatial attention structure from TSN, and full TSM.

As shown in Fig. 13, when we replace AFGAN with TPS, the generated try-on result has artifacts and loses many garment details. In addition, since the garment warped with TPS is excessive, the pattern of the garment is also deformed unnaturally. After removing the generated semantic map from the input, the result is positive, but there is still some color confusion on the edge of the garment. After removing the arm information from the input (without a synthesis strategy), the arms are generated realistically by the TSN, but the fingers are blurred. When the arms overlap (Fig. 13, row 3), the hands cannot be distinguished effectively. After removing the spatial attention structure from the TSN, some generated samples have the problem of color confusion. However, full TSN solves the above problem; the arms are preserved and generated well; the garment has a convincing alignment effect, and its details are largely preserved.

Table VI shows that except for IS, full TSM has the best performance of all the metrics. This proves that the semantic map has guiding value in the generation of try-on results, that the synthesis strategy can effectively retain the details of the arms, and the spatial attention structure in TSN has more advantages in optimizing the generation effect.

V. DISCUSSION

The task of virtual try-on is a great challenge with different poses and various in-shop garments. The human pose is complex and variable, which may include many crossed states between arms and garments and many overlapping states of arms, as shown in Fig. 11. The style and shape of the in-shop garment is unpredictable, and different types of sleeves and collars can make the task more difficult. The garment collar type is also a challenge after fitting in the task. The most important issue is the preservation of the garment detail after fitting. The fundamental reason is that CNN cannot handle the larger spatial deformation, and the rough geometry matching algorithm (e.g., TPS) cannot reasonably focus on the characteristics of the garment. The challenges also include the impact of scene change, the recovery of arms and other body parts, etc. Our proposed method predicts the semantic map to maintain the original shape of the collar and sleeve as correctly as possible. It can strictly constrain the generation of the try-on result. Then, the dense appearance flow is predicted to deform the in-shop garment. Among them, the predicted semantic map is crucial to the constraint of the shape of the garment to be deformed. It can make the alignment of the garment with the body natural and reasonable. In addition, the dense appearance flow can also learn the corresponding characteristic changes after the garment is deformed. Moreover, our arm-handling strategy also greatly preserves the original features of the human body. However, the garment deformation and synthesis still need to be improved due to the variability of scene and posture.

In addition, non-frontal garment images in the dataset may cause the errors in garment alignment. The frequent labelling error in the dataset may also cause the failure of the strategy to prevent arm occlusion and even try-on results. A highly generalizable virtual try-on network is developed by a scenario-rich dataset with a wide variety of body poses and garment styles. It is also essential to ensure that all annotations of the dataset are correct. If the application is used by the mobile device, it will be necessary to develop a lightweight model to implement real-time computing. The generated result is based on more auxiliary condition information in current works, and the more efficient model should be studied with as little input information as possible. Consequently, there are still gaps and unforeseen difficulties between the virtual and real world fields. More work needs to be done to explore the data and implementation schemes of virtual try-ons in different fields.

VI. CONCLUSION

In this work, we propose an image-based virtual try-on network based on the semantic constraints and flow alignment, VTON-SCFA, which retains many details of the human body and garment in the generated result. The GLSP is designed to

predict the exact target semantic map, and the arm representation map is introduced to prevent the arms from being covered by the garment. GLB is designed to analyze the relationship between global and local condition information effectively. A novel garment alignment network, AFGAN, is proposed to effectively align the target garment with the human body, and flow correction is introduced to restrict appearance flow to prevent unnatural warping, and a region constraint is introduced to remove redundant pixels to prevent the arm occlusion and pixel confusion. Finally, the synthesis strategy is used to preserve maximum arms and other body parts. Experiments demonstrate that compared with the state-of-the-art frameworks, the proposed method achieves superior performances of the quantitative index and the realistic visual effect on the virtual try-on benchmark. In future works, we will expand our framework to study the effect of multipose virtual try-on.

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